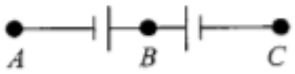
# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

A potentiometer is connected across $A$ and $B$ and a balance is obtained at $64.0$ cm. When potentiometer lead to $B$ is moved to $C$, a balance is found at $8.0$ cm. If the potentiometer is now connected across $B$ and $C$, a balance will be found at




**(A)** $8.0$ cm

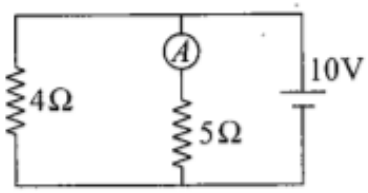
**(B)** $56.0$ cm

**(C)** $64.0$ cm

**(D)** $72.0$ cm

**Q.2**

In the circuit shown in Fig., the reading of the ammeter is (assume internal resistance of the battery be to zero)




**(A)** $\frac{40}{29}$ A

**(B)** $\frac{10}{9}$ A

**(C)** $\frac{5}{3}$ A

**(D)** $2$ A

**Q.3**

The length of a wire of a potentiometer is $100$ cm, and the e.m.f. of its standard cell is $E$ volt. It is employed to measure the e.m.f. of a battery whose internal resistance is $0.5\Omega$. If the balance point is obtained at $l = 30$ cm from the positive end, e.m.f. of the battery is
where $i$ is the current in the potentiometer wire.

**(A)** $\frac{30E}{100}$

**(B)** $\frac{30E}{100.5}$

**(C)** $\frac{30E}{(100-0.5)}$

**(D)** $\frac{30(E-0.5i)}{100}$

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**Q.4**

In a metre bridge experiment, null point is obtained at 20 $\mathrm{cm}$ from one end of the wire when resistance $X$ is balanced against another resistance $Y$. If $X < Y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of $4X$ against $Y$ ?

**(A)** $50\ \mathrm{cm}$

**(B)** $80\ \mathrm{cm}$

**(C)** $40\ \mathrm{cm}$

**(D)** $70\ \mathrm{cm}$

**Q.5**

In a potentiometer experiment, the balancing with a cell is at length $240\ \mathrm{cm}$. On shunting the cell with a resistance of $2\Omega$, the balancing length becomes $120\ \mathrm{cm}$. The internal resistance of the cell is

**(A)** $2\Omega$

**(B)** $4\Omega$

**(C)** $0.5\Omega$

**(D)** $1\Omega$

**Q.6**

The resistance of a galvanometer is $10\Omega$. It gives full-scale deflection when $1\ \mathrm{mA}$ current is passed. The resistance connected in series for converting it into a voltmeter of $2.5\ \mathrm{V}$ will be

**(A)** $24.9\Omega$

**(B)** $249\Omega$

**(C)** $2490\Omega$

**(D)** $24900\Omega$

**Q.7**

A milliammeter of range $10\ \mathrm{mA}$ has a coil of resistance 1 $\Omega$. To use it as an ammeter of range $1\ \mathrm{A}$, the required shunt must have a resistance of

**(A)** $\frac{1}{101}\Omega$

**(B)** $\frac{1}{100}\Omega$

**(C)** $\frac{1}{99}\Omega$

**(D)** $\frac{1}{9}\Omega$

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**Q.8**

In an experiment to measure the internal resistance of a cell by a potentiometer, it is found that the balance point is at a length of $2\ \mathrm{m}$ when the cell is shunted by a $5\Omega$ resistance and is at a length of $3\ \mathrm{m}$ when the cell is shunted by a $10\Omega$ resistance, the internal resistance of the cell is then

**(A)** $1.5\Omega$

**(B)** $10\Omega$

**(C)** $15\Omega$

**(D)** $1\Omega$

**Q.9**

A voltmeter has a resistance of $G$ ohm and range $V$ volt. The value of resistance used in series to convert it into voltmeter of range $nV$ volt is

**(A)** $nG$

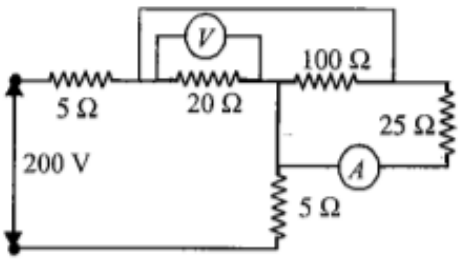
**(B)** $(n-1)G$

**(C)** $G/n$

**(D)** $G/(n-1)$

**Q.10**

In the figure, voltmeter and ammeter shown are ideal. Then voltmeter and ammeter readings, respectively, are




**(A)** $125\ \mathrm{V}, 3\ \mathrm{A}$

**(B)** $100\ \mathrm{V}, 4\ \mathrm{A}$

**(C)** $120\ \mathrm{V}, 4\ \mathrm{A}$

**(D)** $120\ \mathrm{V}, 3\ \mathrm{A}$

**Q.11**

The operating temperature of the filament of lamp is $2000^{\circ}\mathrm{C}$. The temperature coefficient of the material of filament is $0.005^{\circ}\mathrm{C}^{-1}$. If the atmospheric temperature is $0^{\circ}\mathrm{C}$, then the current in the $100\ \mathrm{W} - 200\ \mathrm{V}$ lamp when it is switched on is nearest to

**(A)** $2.5\ \mathrm{A}$

**(B)** $3.5\ \mathrm{A}$

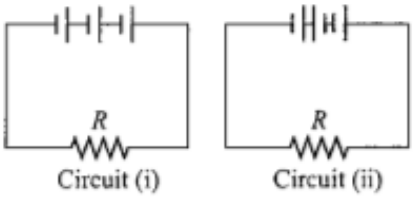
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**(C)** $4.5\ \text{A}$

**(D)** $5.5\ \text{A}$

**Q.12**

Three identical cells, each having an e.m.f. of $1.5\ \text{V}$ and a constant internal resistance of $2.0\Omega$, are connected in series with a $4.0\Omega$ resistor $R$, first as in circuit (i), and secondly as in circuit (ii).




Circuit (i) Circuit (ii)

What is the ratio $\frac{\text{Power in } R \text{ in circuit (i)}}{\text{Power in } R \text{ in circuit (ii)}}$ ?

**(A)** $9.0$

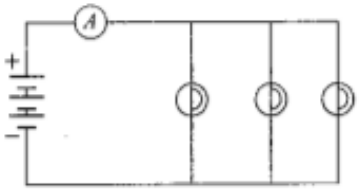
**(B)** $7.2$

**(C)** $1.8$

**(D)** $3.0$

**Q.13**

Three similar light bulbs are connected to a constant voltage d.c. supply as shown in Fig. Each bulb operates at normal brightness and the ammeter (of negligible resistance) registers a steady current.




The filament of one of the bulbs breaks. What happens to the ammeter reading and to the brightness of the remaining bulbs?

Ammeter reading & Bulb brightness

**(A)** increases, increases

**(B)** increases, unchanged

**(C)** unchanged, unchanged

**(D)** decreases, unchanged

**Q.14**

Two identical batteries each of e.m.f. $E = 2\ \text{V}$ and internal resistance $r = 1\Omega$ are available to produce heat

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in an external circuit. What is the maximum rate of production of heat that can be obtained in the external circuit?

**(A)** $1\ \mathrm{W}$

**(B)** $2\ \mathrm{W}$

**(C)** $4\ \mathrm{W}$

**(D)** $8\ \mathrm{W}$

**Q.15**

Two similar head light lamps are connected in parallel to each other. Together, they consume $48\ \mathrm{W}$ from a $6\ \mathrm{V}$ battery. What is the resistance of each filament?

**(A)** $6\Omega$

**(B)** $4\Omega$

**(C)** $3.0\Omega$

**(D)** $1.5\Omega$

**Q.16**

Two electric bulbs, rated for the same voltage, have powers of 200 and $100\ \mathrm{W}$, respectively. If their resistances are $r_1$ and $r_2$, respectively, then

**(A)** $r_1 = 2r_3$

**(B)** $r_2 = 2r_1$

**(C)** $r_2 = 4r_1$

**(D)** $r_1 = 4r_2$

**Q.17**

An electric bulb rated for $500\ \mathrm{W}$ at $100\ \mathrm{V}$ is used in a circuit having a $200\ \mathrm{V}$ supply. The resistance $R$ that must be put in series with the bulb, so that the bulb draws $500\ \mathrm{W}$, is

**(A)** $18\Omega$

**(B)** $20\Omega$

**(C)** $40\Omega$

**(D)** $700\Omega$

**Q.18**

$n$ identical light bulbs, each designed to draw $p$ power from a certain voltage supply, are joined in series across that supply. The total power which they will draw is

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**(A)** $nP$

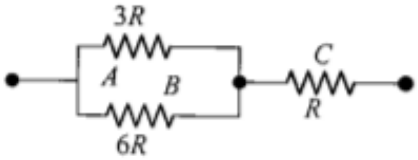
**(B)** $P$

**(C)** $P/n$

**(D)** $P/n^2$

**Q.19**

Figure shows a network of three resistances. When some potential difference is applied across the network, thermal powers dissipated by $A, B$ and $C$ are in the ratio




**(A)** $2:3:4$

**(B)** $2:4:3$

**(C)** $4:2:3.$

**(D)** $3:2:4$

**Q.20**

Two identical heaters rated $220\ \text{V} - 1000\ \text{W}$ are placed in series with each other across $220\ \text{V}$ line; then the combined power is

**(A)** $1000\ \text{W}$

**(B)** $2000\ \text{W}$

**(C)** $500\ \text{W}$

**(D)** $4000\ \text{W}$

**Q.21**

Consider a wire of length $l$, area of cross section A and resistivity $\rho$ with resistance $10\Omega$. Its length is increased by applying a force and it becomes four times of its original value. Find the changed resistance of the wire.(in ohm)

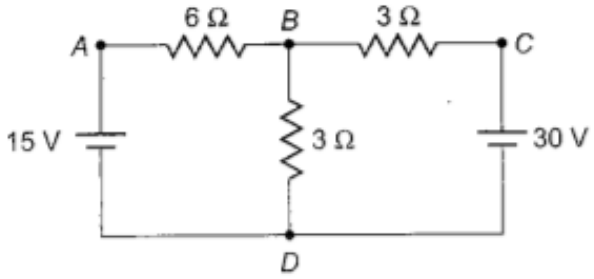
**Q.22**

A wire of mass $m$, length $l$, density $d$, and area of cross section $A$ is stretched in such a way that its length increases by $10\%$ of its original value. Express the changed resistance in percentage. (in %)

**Q.23**

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In the circuit shown in Fig., find the current through the branch $BD$.

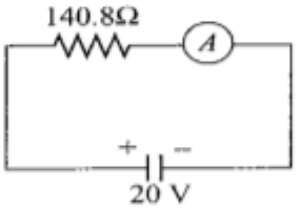



**Q.24**

A potentiometer wire of length $10\ \mathrm{m}$ and resistance $30\Omega$ is connected in series with a battery of e.m.f. $2.5\ \mathrm{V}$, internal resistance $5\Omega$ and an external resistance $R$. If the fall of potential along the potentiometer wire is $50\mu\mathrm{V/mm}$, then the value of $R$ is found to be $23n\Omega$. What is $n$ ?

**Q.25**

The ammeter shown in the figure. consists of $480\Omega$ coil connected in parallel to $20\Omega$ shunt. The reading of the ammeter comes out to be $\frac{1}{*}\ '\ \mathrm{A}$ : What is '*'?




**Q.26**

A galvanometer together with an unknown resistance in series is connected across two identical batteries of each $1.5\ \mathrm{V}$. When the batteries are connected in series, the galvanometer records a current of $1\ \mathrm{A}$ and when the batteries are connected in parallel, the current is $0.6\ \mathrm{A}$. In this case, internal resistance of the battery is $\frac{1}{*}\Omega$ . What is the value of '*'?

**Q.27**

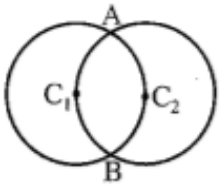
A $5\ \mathrm{m}$ potentiometer wire having $3\Omega$ resistance per metre is connected to a storage cell of steady e.m.f. $2\ \mathrm{V}$ and internal resistance $1\Omega$. A primary cell is balanced against $3.5\ \mathrm{m}$ of it. When a resistance of $32/n\Omega$ is put in series with the storage cell, the null point shifts to the centre of the last wire, i.e., $4.5\ \mathrm{m}$. What is ' $n$ '?

**Q.28**

Two circular rings of identical radii and resistance of $36\Omega$ each are placed in such a way that they cross each other's centre $C_1$ and $C_2$ as shown in the figure. Conducting joints are made at intersection points $A$ and $B$ of the rings. An ideal cell of e.m.f. $20\ \mathrm{V}$ is connected across $A$ and $B$. Find the power delivered by cell (in

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$10^2$ W).




**Q.29**

A heating coil is rated $100$ W, $220$ V. The coil is cut in half and two pieces are joined in parallel to the same source. Now what is the energy (in $\times 10^2$ J ) liberated per second?

**Q.30**

A dynamo develops $0.5$ A at $6$ V. Find the energy which it generates in $1$sec.

## Answers & Solutions

**Q.1 Answer:**

$56.0$ cm

**Solution:**

$E_1 \propto 64$
$E_1 - E_2 \propto 8$
$E_2 \propto l$
$\therefore \quad 64 - l = 8$
**or** $l = 64 - 8 = 56$ cm

**Q.2 Answer:**

$2$ A

**Solution:**

**Voltage across** $5\Omega = 10$ V
$\therefore \quad I = \frac{10}{5}$ A $= 2$ A

**Q.3 Answer:**

$\frac{30E}{100}$

**Solution:**

**Using the principle of potentiometer,** $V \propto l \therefore \quad \frac{V}{E} = \frac{l}{L}$ **or** $V = \frac{l}{L}E = \frac{30}{100}E = \frac{30E}{100}$

**Q.4 Answer:**

$50$ cm

**Solution:**

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$\frac{X}{Y} = \frac{20}{80} = \frac{1}{4}$ or $Y = 4X$
$\frac{4X}{Y} = \frac{l}{100-l}$ or $\frac{4X}{4X} = \frac{l}{100-l}$ or $l = 50$ cm

**Q.5 Answer:**

$2\Omega$

**Solution:**

$r = \frac{l_1 - l_2}{l_2} R = \frac{240-120}{120} \times 2\Omega = 2\Omega$

**Q.6 Answer:**

$2490\Omega$

**Solution:**

$i_z = \frac{V}{G+R} \Rightarrow 10^{-3} = \frac{2.5}{10+R} \Rightarrow R = 2490\Omega$

**Q.7 Answer:**

$\frac{1}{99}\Omega$

**Solution:**

$S = \frac{i_g G}{(I - i_g)} = \frac{0.01 \times 1}{1 - 0.01} = \frac{1}{99}\Omega$

**Q.8 Answer:**

$10\Omega$

**Solution:**

In case of internal resistance measurement by potentiometer,
$\frac{V_1}{V_2} = \frac{\ell_1}{\ell_2} = \frac{[ER_1/(R_1+r)]}{[ER_2/(R_2+r)]} = \frac{R_1(R_2+r)}{R_2(R_1+r)}$
Here $\ell_1 = 2$ m, $\ell_2 = 3$ m, $R_1 = 5\Omega$ and $R_2 = 10\Omega$
$\therefore \quad \frac{2}{3} = \frac{5(10+r)}{10(5+r)}$ or $r = 10\Omega$

**Q.9 Answer:**

$(n-1)G$

**Solution:**

We know that $R = \frac{V}{I_g} - G$
The voltmeter gives the full-scale deflection for potential difference $V$. Its resistance is $G$. Hence, $I_n = (V/G)$.
Given that $V = nV$
$\therefore \quad R = \frac{nV}{(V/G)} - G = (n-1)G$

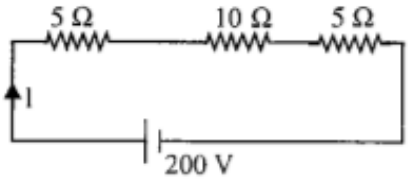
**Q.10 Answer:**

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$100\ \text{V}, 4\ \text{A}$

**Solution:**

**Resistors $20\Omega, 100\Omega$ and $25\Omega$ will be in parallel. Their equivalent is $10\Omega$.**




$I = \frac{200}{5+10+5} = 10\ \text{A}$

**p.d. across $10\Omega$,** $10I = 10 \times 10 = 100\ \text{V}$

**This will be the voltmeter reading. Also, this will be the p.d. across each of $20\Omega, 100\Omega$ and $25\Omega$ resistors.**

**Ammeter reading = current through $25\Omega = \frac{100}{25} = 4\ \text{A}$.**

**Q.11 Answer:**

$5.5\ \text{A}$

**Solution:**

$R_{200} = \frac{200\times200}{100} = 400\Omega$

**So,** $400 = R_0[1 + 0.005 \times 2000]$

$\therefore R_0 = \frac{400}{11}\Omega \approx 36\Omega$, **Hence, current** $I = \frac{200}{R_0} = 5.5\ \text{A}$

**Q.12 Answer:**

$9.0$

**Solution:**

**Power** $P = I^2R = \left(\frac{V}{R+3r}\right)^2 R \Rightarrow P \propto V^2$

**Power ratio** $= \left(\frac{V_1}{V_2}\right)^2 = \left(\frac{4.5}{1.5}\right)^2 = 3^2 = 9$

**Q.13 Answer:**

decreases, unchanged

**Solution:**

**Suppose $V$ is the voltage of the supply and $R$ is the resistance of each bulb.**

**Now, $R_p = R/3$ and current in ammeter, $I = V/R_p = 3\ \text{V}/R$, provided all three bulbs are working properly.**

**If one bulb has broken down, then**

$R_p = \frac{R}{2}$ and $I = 2\frac{V}{R}$

**Therefore, current decreases and since current through each bulb is $V/R$ the same as before, brightness of bulbs is not affected.**

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**Q.14 Answer:**

$2\ \mathrm{W}$

**Solution:**

**Let both the batteries are connected in series. Power output is maximum when external resistance is $2\Omega$. Current in the circuit $= 4\ \mathrm{V}/4\Omega = 1\ \mathrm{A}$ and power in the external circuit $= (1)^2 \times 2 = 2\ \mathrm{W}$**

**Q.15 Answer:**

$1.5\Omega$

**Solution:**

**Power consumed by each lamp $= 24\ \mathrm{W}$. Hence using $R = \left(V^2/P\right)$ we find**
$R = (36/24) = 1.5\Omega$

**Q.16 Answer:**

$r_2 = 2r_1$

**Solution:**

$P = \frac{V^2}{R}, \frac{P_1}{P_2} = \frac{R_2}{R_1} \Rightarrow r_2 = 2r_1$

**Q.17 Answer:**

$20\Omega$

**Solution:**

$P = VI, I = \frac{P}{V}$ **or** $I = \frac{500\ \mathrm{W}}{100\ \mathrm{V}} = 5\ \mathrm{A}$



**Now,** $5R = 100$ **or** $R = 20\Omega$

**Q.18 Answer:**

$P/n$

**Solution:**

**Total resistance** $= \frac{nV^2}{P}$**. Power** $= \frac{V^2 P}{nV^2} = \frac{P}{n}$

**Q.19 Answer:**

$4 : 2 : 3.$

**Solution:**

**Let current flow from $b$ to $a$ as shown (Fig.).**
**Ratio is** $\left(\frac{2}{3}I\right)^2 3R : \left(\frac{1}{3}I\right)^2 6R : I^2 R$

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**or** $\frac{4}{3} : \frac{2}{3} : 1$ **or** $4 : 2 : 3$

**Q.20 Answer:**

$500\ \text{W}$

**Solution:**

**Let the resistance of the two heaters be denoted by** $R_1$ **and** $R_2$**, respectively. Then**

$R_1 = \left(\frac{V^2}{P_1}\right)$ **and** $R_2 = \left(\frac{V^2}{P_2}\right)$

**If the resistance of the series combination is denoted by** $R_s$ **and the corresponding power by** $R_s$**, then** $R_s = R_1 + R_2$**.**

**or** $\frac{V^2}{P_S} = \frac{V^2}{P_1} + \frac{V^2}{P_2}$

**or** $P_s = \frac{P_1 P_2}{P_1 + P_2} = \frac{1000 \times 1000}{2000} = 500\ \text{W}$

**Q.21 Answer:**

—

**Solution:**

**Here** $l_1 = l$, $A_1 = A$, **and** $R = 10\Omega$**. Similarly,** $l_2 = 4l$ **and** $R_2 =$ **? Resistivity is same in each case as the material is same. The volume of the wire will remain the same even after the increase in the length.** $A_1 l_1 = A_2 l_2 \Rightarrow A_2 = \frac{A_1 l_1}{l_2} = \frac{Al}{4l} = \frac{A}{4}$

**The formula used for measuring resistance of wire is** $R = \rho(l/A)$**.**

**Using this formula in both cases, we have**

$R_1 = \rho\frac{l_1}{A_1} = \frac{\rho l}{A} \quad \ldots(\text{i})$

$R_2 = \rho\frac{l_2}{A_2} = \rho\frac{4l}{A/4} = 16\rho\frac{l}{A} \quad \ldots(\text{ii})$

**Dividing Eq. (ii) by Eq. (i),** $\frac{R_2}{R_1} = 16 \Rightarrow R_2 = 160\Omega$

**Q.22 Answer:**

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**Solution:**

**Given mass** $m$**, length** $l_1 = l$**, density** $d$**, and area of cross section** $A_1 = A$**. Let** $\rho$ **be the resistivity and** $R_1$ **be the resistance of the wire. Mass of wire** $m =$ **volume** $\times$ **density** $= Al \times d = Ald$**. Therefore, area of cross section** $A_1 = m/ld$**, the resistance of the wire is**

$R_1 = \rho\frac{l}{A_1} = \rho\frac{l^2 d}{m} = \left(\frac{\rho d}{m}\right) l^2 = kl^2$ **(i)**

**Let** $l_2$ **be the new length, then** $l_2 = l + \frac{10}{100}l = l + 0.1l = 1.1$

**Let** $R_2$ **be the resistance of the wire after stretching, then**

$R_2 = kl_2^2$ **(ii)**

**Dividing Eq. (ii) by Eq. (i), we have** $\frac{R_2}{R_1} = \frac{l_2^2}{l_1^2} = \frac{(1.1)^2 l_1^2}{l_1^2} = 1.21$

$\Rightarrow \quad R_2 = 1.21R_1 = R_1 + 0.21R_1 \Rightarrow R_2 - R_1 = 0.21R_1$

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**Hence, the percentage change in the resistance is**
$\frac{R_2-R_1}{R_1} \times 100 = 21\%$

**Q.23 Answer:**

—

**Solution:**

**To find the current in a particular branch in a circuit where resistances and voltages are given, lower loop rule is applicable. In order to find out the solution, the given circuit is directed into parts or close loops which are analyzed using Kirchhoff's second law and corresponding sign conventions. Assume the currents in the circuit as shown in Fig.**
**Applying** $\mathrm{KVL}$ **along the loop** $ABDA$ **and moving in clockwise direction, we get**
$-6I_1 - 3I_2 + 15 = 0 \Rightarrow 2I_1 + I_2 = 5$ **(i)**
**While moving in the direction of current corresponding IR, products are taken as negative and if the negative terminal of the battery comes first, then e.m.f. is taken as +ve (Fig.).**


**Applying KVL along the loop** $BCDB$**, we get**
$-3\left(I_1 - I_2\right) - 30 + 3I_2 = 0 \Rightarrow -I_1 + 2I_2 = 10$ **(ii)**
**Solving Eqs. (i) and (ii) for** $I_2$**, we get** $I_2 = 5\ \mathrm{A}$.

**Q.24 Answer:**

—

**Solution:**

**Potential gradient,** $k = \frac{IR_P}{L} = \frac{ER_p}{(R+R_p+r)L}$
**(where** $R_p$ **is resistance of the wire)**
$\Rightarrow 50 \times 10^{-3} = \frac{2.5 \times 30}{(R+30+5) \times 10} \Rightarrow R = 115\Omega$

**Q.25 Answer:**

—

**Solution:**

**Equivalent resistance of the ammeter is**
$\frac{(480\Omega)(20\Omega)}{480\Omega+20\Omega} = 19.2\Omega$
**The equivalent resistance of the circuit is** $140.8\Omega + 19.2\Omega = 160\Omega$**. Therefore, current,** $i = 20\ \mathrm{V}/160\Omega = (1/8)\mathrm{A}$

**Q.26 Answer:**

—

**Solution:**

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Let $R$ be the combined resistance of galvanometer and an unknown resistance and $r$ the internal resistance of each battery. When the batteries, each of e.m.f. $E$ are connected in series, the net e.m.f. $= 2E$ and net internal resistance $= 2r$.

Current $i_1 = \frac{2E}{R+2r} \Rightarrow 1.0 = \frac{2\times1.5}{R+2r}$

$\Rightarrow \quad R + 2r = 3$ (i)

When the batteries are connected in parallel, the e.m.f. remains $E$ and net internal resistance become $r/2$. Therefore,

$$\text{current is} \quad i_2 = \frac{E}{R+\frac{r}{2}} = \frac{2E}{2R+r}$$

$$\Rightarrow \quad 2R + r = \frac{2E}{i_2} = \frac{2\times1.5}{0.6} = 5.0 \quad ...(\text{ii})$$

Solving Eqs. (i) and (ii), we get $r = \frac{1}{3}\Omega$

**Q.27 Answer:**

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**Solution:**

This means you will not get the balancing point.

$$i = \frac{2}{1+15} = \frac{1}{8}, E_P = \frac{1}{8} \times 3 \times \frac{7}{2} = \frac{21}{16}$$

$$\frac{21}{16} = i \times 3 \times \frac{9}{2} \Rightarrow i = \frac{7}{72}\ \text{A},\ \text{now}\ \frac{7}{72} = \frac{2}{16+R} \Rightarrow R = \frac{32}{7}\Omega$$

**Q.28 Answer:**

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**Solution:**

The resistance of four sections are shown in the figure.



Hence, the equivalent resistance $R$ across $AB$ is

$$\frac{1}{R} = \frac{1}{24} + \frac{1}{12} + \frac{1}{12} + \frac{1}{24} \Rightarrow R = 4\Omega$$

$$\text{Power} = V^2/R = 20^2/4 = 100\ \text{W}$$

**Q.29 Answer:**

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**Solution:**

Let resistance of heating coil be $R$, then

$$100 = \frac{220^2}{R} \Rightarrow R = \frac{(220)^2}{100}\Omega$$

Resistance of each cut part $= R/2$

$$\text{Now, power dissipated} = P = 2\left(\frac{(220)^2}{R/2}\right)$$

$$\Rightarrow \quad P = 4\frac{(220)^2}{R} = 4 \times 100 = 400\ \text{W}$$

**Q.30 Answer:**

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**Solution:**

$R = \frac{6}{0.5} = 12\Omega$
$H = I^2Rt = (0.5)^2 \times 12 \times 1 = 3 \text{ J}$

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